



Software Testing

CHAPTER 3. SOFTWARE TESTING METHODS AND TECHNIQUES

Session 2:

SOFTWARE TESTING APPROACHES



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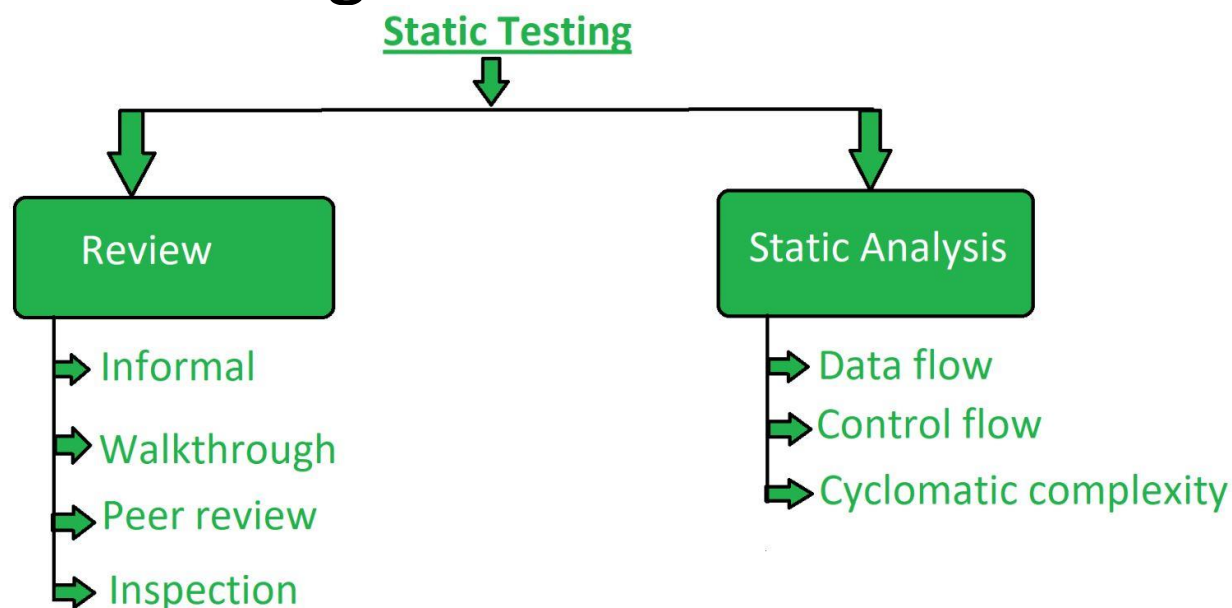
1. Static testing

- ❖ Is a type of software testing method, which is used to check defects **without executing code.**
- ❖ The primary goal of Static Testing is **to catch defects in the documentation phase**, which helps prevent error at later stages and is therefore cost-effective.



1. Static testing

❖ There are mainly two type techniques used in Static Testing:





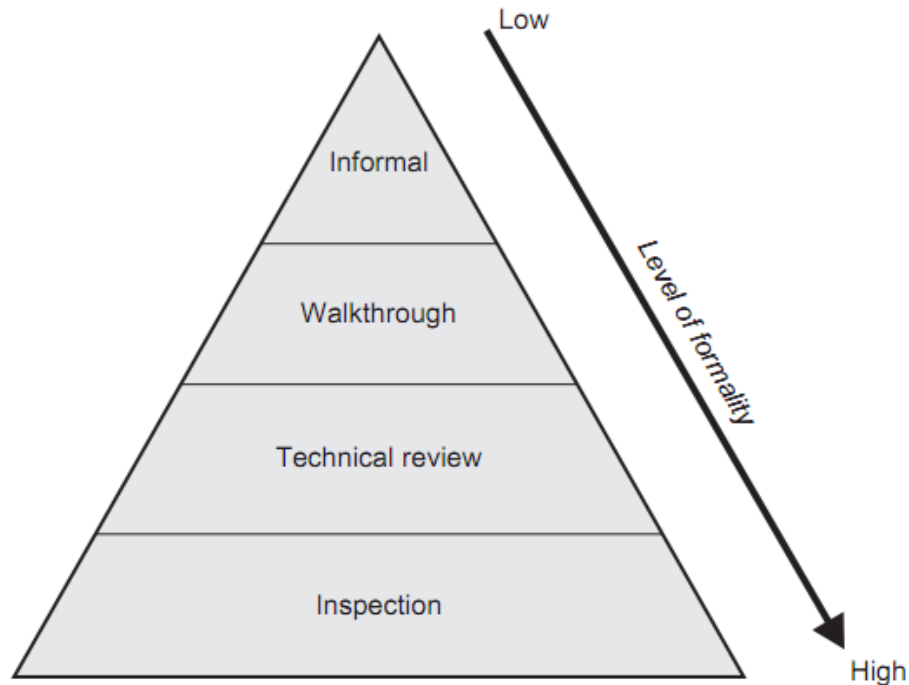
1.1 Review

- ❖ Review is a process or technique that is performed to **find the potential defects in the design of the software.**
- ❖ Review is performed to identify and correct errors found in documents such as requirements, design, implementation, test cases, and maintenance....



1.1 Review

❖ Type of review: The different levels of formality by review type





1.1 Review

❖ Informal review

- Informal reviews are applied in the early stages of the life cycle of the document.
- These reviews are conducted between two person. The main purpose is for “discussion”
- These reviews are not based on the procedure and **not documented.**



1.1 Review

❖ Walkthrough

- Is a formal review process which usually happens in the meeting room
- It is **led by the authors**
- Author guide the participants through the document according to his or her thought process to **achieve a common understanding and to gather feedback.**

⇒ The goal is:

- ✓ Finding defects
- ✓ help team members gain an understanding of the content of the document



1.1 Review

❖ Technical review (Peer review)

- Technical reviews are **documented**
- Technical review is a discussion meeting that focuses on technical content of the document
- **Defects are found by the experts** (such as architects, designers, key users) who focus on the content of the document.

=>The goal is to ensure the value of technical concept are used correctly.



1.1 Review

❖ Inspection:

- It is the **most formal review technique**
- It is led by the trained moderators
- It is review of business requirements, functional requirements, system design, code, testing activities,...
- Inspection has defined entry and exit criteria for product which is under review.

⇒ The goal is:

- ✓ Finding defects as early as possible
- ✓ Improve the quality of the document under inspection
- ✓ Reports or documents are maintained and kept for future records.



1.1 Review

❖ Inspection review process (formal review):





1.2 Static analysis

- ❖ Performed on requirement design or code without actually executing the software or before the code is actually run
- ❖ The goal is to **find defects in software source code** and software models.
- ❖ Typical defects discovered by static analysis tools include:
 - Unused variables
 - Dead code (or unreachable code)
 - Infinite loops
 - Variable with undefined value
 - Security vulnerabilities
 - Syntax violations of code and software models



1.2 Static analysis

❖ Benefits of static analysis:

- Early detection of defects prior to test execution.
- Early warning about suspicious aspects of the code or design, by the calculation of metrics, such as a high-complexity measure.
- Identification of defects not easily found by dynamic testing
- Improved maintainability of code and design.
- Prevention of defects



1.2 Static analysis

❖ Static Analysis is of three types:

- Data Flow:
 - ✓ Data flow is related to the stream processing.
- Control Flow:
 - ✓ Control flow is basically how the statements or instructions are executed.
- Cyclomatic Complexity:
 - ✓ Cyclomatic complexity is the measurement of the complexity of the program that is basically related to the number of independent paths in the control flow graph of the program.



2. Dynamic testing

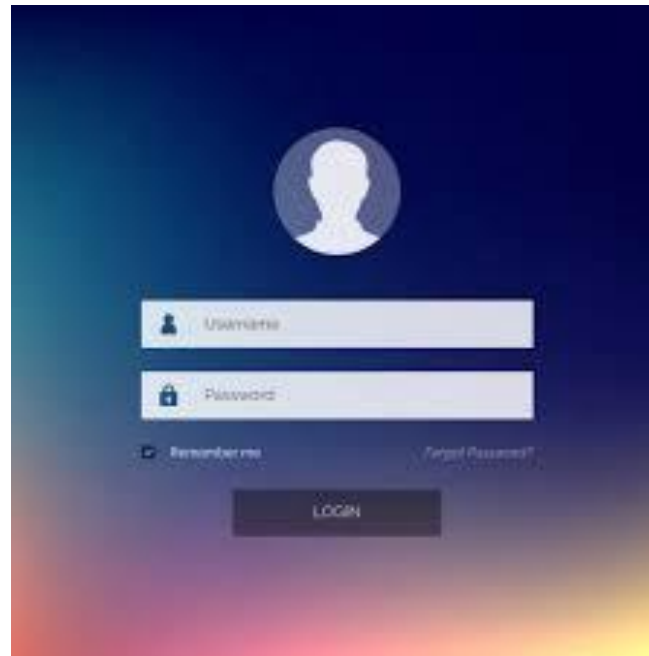
- ❖ Dynamic Testing is the type of testing that validates the functionality of an application by **executing the code**.
- ❖ Testing the application by giving input values and validate the expected output with the actual output.
- ❖ Dynamic testing is divided into two categories:
 - White box testing
 - Black box testing



2. Dynamic testing

❖ Example: Testing login functionality

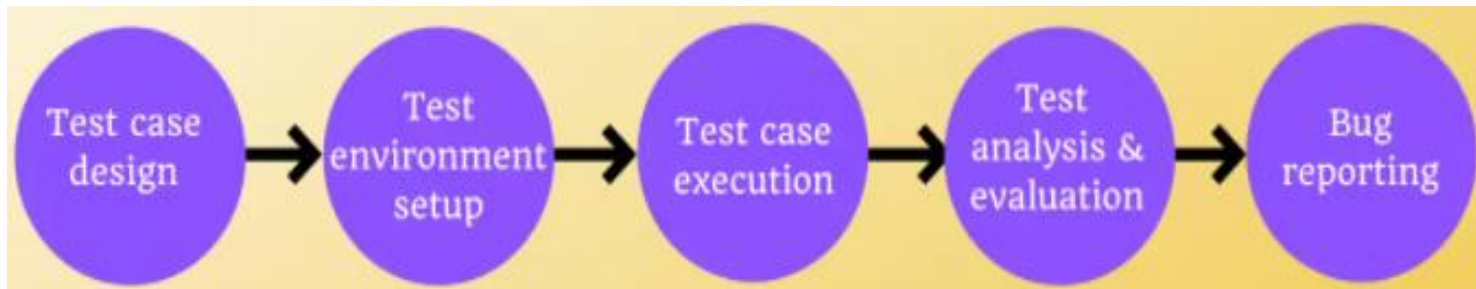
- Require: the Username is restricted to Alphanumeric.





2. Dynamic testing

❖ Dynamic Testing Process:





2. Dynamic testing process

❖ Step 1: Test case design

- ✓ Design test cases based on the requirements.
- ✓ Identify features to be tested
- ✓ Derive the Test Conditions
- ✓ Derive the coverage Items

❖ Step 2: Environment Setup

- ✓ install the build and manage the test machines.
- ✓ Make sure that the Testing Environment is always comparable to the Production environment.

❖ Step 3: Test Execution

- ✓ Test case are actually executed

❖ Step 4: Test Analysis and Evaluation

- ✓ Analyse and evaluate the findings generated from the testing by comparing the actual results with the expected results

❖ Step 5: Bug reporting

- ✓ If the Actual Results and Expected results are not same then the Test case has to be marked as Fail
- ✓ A Bug should be logged.



Static testing vs Dynamic testing

Static testing	Dynamic testing
It involves examining work products without executing the code	It involves the execution the code
Finds defects	Finds the failures in the system
Aims at finding defects early in the life cycle	Is carried out during and after the development phase
Rework cost is relatively low	Rework cost is relatively high
Includes the verification process	Includes the validation process
Assesses the documentation and code	Finds the bugs in the software system



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Thank You !